

Development of Sign Language using Flex Sensors

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Abstract—Sign Language is not familiar to those who are not deaf or dumb. When humans talk to each other, they convey their words through both speech and gestures. For a deaf and dumb individual, it is really difficult to lead a normal life with such a big communication barrier. This affects their social life as well as relationships. So, this framework is developed to assist them and help them in conveying their message easily. The creation of such a device involves both electronics and computer engineering knowledge. It will include flex sensors, Arduino, Bluetooth module, accelerometer, glove and an android application to display the results. The framework plans to reduce the communication barrier between a normal individual and a deaf/dumb individual. Also, the aim is to realize the gaps and options available in this field to further improve the framework.

Keywords—Flex Sensor, Sign Language, Arduino, Gyroscope, Gesture

I. INTRODUCTION

Around 70 million hard of hearing, quiet, individuals are around the globe as per the World Federation of the Deaf (WFD). These individuals do not feel like sharing their thoughts, have callings and have simple approaches to impart in society; in this way, the requirement for building up a functional answer for these individuals. An undeniable arrangement might be perusing lips which are not a simple method to impart.

Figure 1 shows the different signs for each alphabet. Except, a gesture-based conversation is essential on account that perusing lips isn't always talented due to the fact several phrases inside the English language are homophones and numerous letters are articulated making use of a comparable mouth form. Folks that are tragically challenged frequently will in trendy experience awkward around others, whilst causing to note their hearing difficulty. These individuals want to resemble their partners with remarkable hearing, this force a concept in them to more often than not thoughts their personal enterprise and to not partake in sporting events with those normal individuals.

Gesture-based communications are utilized to skip in contemplation with photos, and articles and so forth. They likewise bypass on the combo of words and pics (for instance motions). Signals are various examples made by way of the twists and twists of the arms. It is clear that now not all people can realize the snapshots and symptoms utilized by the hard of hearing/quiet individuals. Henceforth, the development of gadgets for expertise hand trends. In this structure, flex resistors were utilized to quantify the electrical voltage applied with each signal, and the letters were created by looking at the qualities. This framework was trailed by numerous others utilizing various advancements, like pressure sensors, contact sensors, spinners and additionally accelerometer.

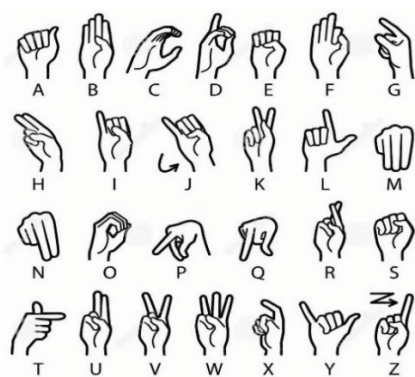


Fig. 1. Sign Language representation by hand

Because of nonattendance of correspondence between a run of the mill individual and in need of a hearing aid and dumb person, the extent is lessening of proficient and utilized tragically challenged.

To beat this problem, a stand-out software is evolved. Our software indicates fill in as a captivating interpreter used to unravels communicate thru signing in form of motion via an idiotic individual to a mixed word which has an equal necessity in gesture-primarily based communication which deciphers an important part, as an android software for giving the substance of indication trade. This will help in easing out the communication between a dumb and a normal person.

II. LITERATURE REVIEW

This structure hopes to reduce the constraints in correspondence with a traditional individual. The critical explanation for the gloves is to develop a monetarily clever structure which will offer a voice to the unvoiced person with the assistance of sensing translator. It indicates that the usage of top-notch gloves correspondence won't be a challenge between gatherings. Usage of sharp glove through a character with insufficiency influences people to make and besides they received have variations in all aspects.[1]

This framework attempts to relate the communication by organizing an adaptable glove senses the client's American Sign Language developments and gives the output in a text or speech form at a screen. The glove based translator incorporates flex sensors that measure different resistance values. The glove's Arduino microcontroller examines the sensor readings to see the banner. Utilizing this contraption, new day speakers of American Sign Language may be able to chat with others in an immediate and pleasing manner. [2]

Sensory glove translator is a framework that is going underneath the magnificence of the expanded communication. Augmentative and Optional Correspondence (AOC) is a conspicuous phase in the development of help institutions for humans with disappointments, especially those with veritable preparation. The sensor here is the main player in

measuring the bend value, position, rotation or movement involved in the hand gesture.

An AOC structure is a blended gathering of sections, which includes the photographs, facilitates, structures, and techniques utilized by human beings to remodel correspondence. Those advances continue jogging from all round law-tech structures (i. e., key changes without any batteries or hardware, for instance, correspondence sheets). [3]

Motion based correspondence prepares for nearly deaf calm people to pass in. Through motion-based correspondence, which is plausible for an almost deaf calm individual without the strategy for acoustic sounds. The factor at the back of this is to broaden a system for seeing the signal-based correspondence, which clothing correspondence among humans with communicating obstacle and not unusual human beings, thus diminishing the correspondence hole among them.[4]

III. PROPOSED ARCHITECTURE

The proposed architecture includes both software and hardware. The block diagram of the proposed system is shown in figure 2. The hardware part includes flex sensors for each finger, accelerometer, Arduino as a primary controller, Bluetooth module and glove.

The flex sensors are the elements used to measure the bend values of the gestures made and take the input. They then convert the signs to their respective resistance values. These resistance values increase with the increase in bend. Also, the input from the gyroscope is taken. It measures the angular bend/ rotation. It is used to find the orientation of fingers. The measured resistance values are then checked upon by the Arduino (microcontroller). If the input value is equal to any of the values already stored in the controller, then the text corresponding to that value is passed on to the android application via the Bluetooth module.

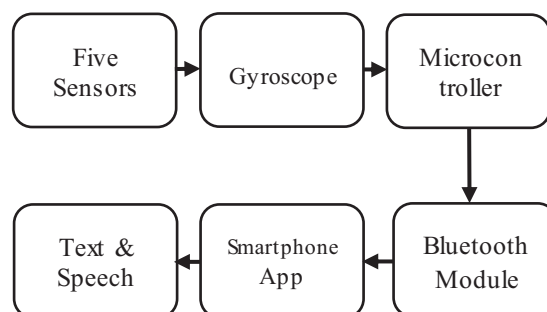


Fig. 2. Sign Language Convertor

The Bluetooth incorporated then transfers the text to the application. Android application, which is a part of a software component, can either be a third-party application or a developed one. The sensor here is the main player in measuring the bend value, position, rotation or movement involved in the hand gesture.

IV. SYSTEM ALGORITHM

The workflow of the project can be explained as-

A. Algorithm

The steps involved are:

Stage 1- Start

Stage 2- Gather the bend values from each of the sensors

Stage 3- If the values match any of the pre-fixed value, then the Bluetooth module will send it forward.

Stage 4- The android application will then display the assigned text on to the screen.

Stage 5- Stop

B. Flowchart

Figure 3 demonstrates the steps involved in the implementation of the proposed architecture. The program starts reading the hand gestures and senses the incoming data. Next, it is fed to the microcontroller where the values are checked. Arduino used as the primary input unit takes the values from sensors as some number. In case the value of the sensor matches any of the fixed values, the controller will pass on the assigned content to the android application via the Bluetooth module. To get new data, these steps get repeated through a loop.

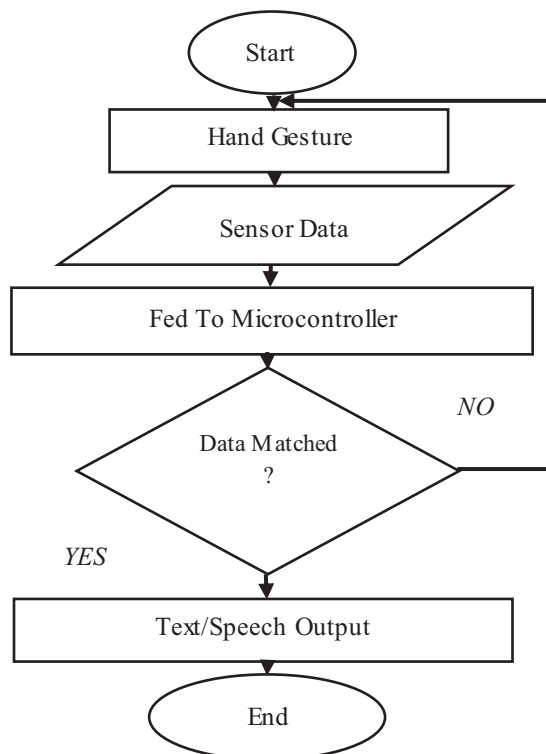


Fig. 3. Flowchart of the framework

V. RESULT

A user can communicate in sign language by simply wearing the glove and making different gestures with fingers. Since each flex sensor has a range of resistances, each sensor was tested apart and its voltage values were taken for different sentences. While testing, the values altered, so the problem was solved by taking different ranges for each flex position.

A. Hardware Model

The hardware involved in this framework includes a flex sensor subsystem, a rotation sensor subsystem (gyroscope) and an output subsystem having an android application. Choosing the right type of sensors for measuring the bend value of the finger joints is a very crucial part. Here, flex sensors have been chosen that are essential elements for taking the input from the user. Figure 4 shows the real equipment model of the proposed framework. It shows five flex sensors mounted on the top of each finger to measure the bend resistance.

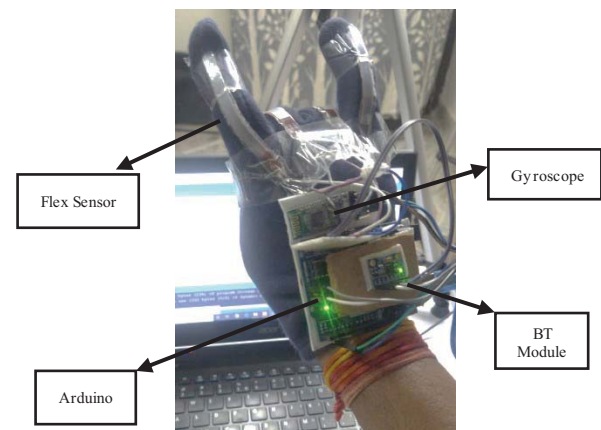


Fig. 4. Hardware model of the project

B. Android Application

The last step of the proposed framework is the android application. This application can be either developed or can be a third-party application. The application can thus be configured for both text and voice or any single of them. The flowchart of the workflow of this application is shown in figure 5:

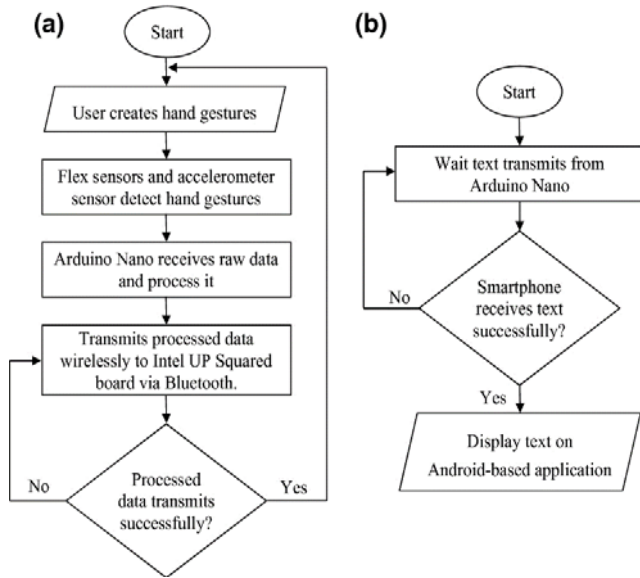


Fig. 5. Android Application Flowchart

Android mobile receives data via Bluetooth in bytes format, convert them into a string. Finally, the string is converted into a voice using the text to speech application of Android mobile. This overall system is mounted over a normal glove for easy handling and recognizes the hand gestures accurately. Besides the text which appears on the screen, the application also gives a voice output of what it is displaying on the screen. This feature is very helpful in case the person is only dumb and not deaf, so he/she can hear if the gesture to sign conversion is right or wrong. The android application's display is shown in figure 6.

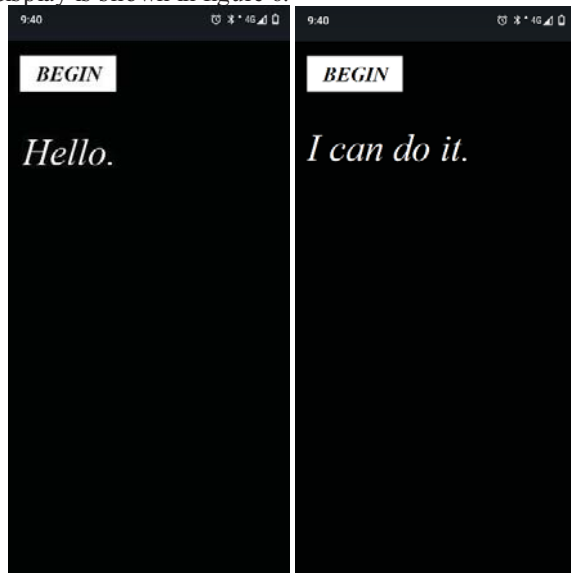


Fig. 6. Android Application displaying the converted text

This application can further be enhanced with the inclusion of facial expressions recognizing sensors or the cameras. What they will do is, they will grasp the facial expressions and will convert them into the text which the user is trying to convey. So, different modes can be set

up in the application. One can include reading the facial expressions too. While the other stays the same involving the use of finger bend values or hand gestures only.

The system proposed in this paper achieved an average accuracy of 92%. Table I shows the accuracy and response time of some sentences corresponding to their signs.

TABLE I. ACCURACY AND RESPONSE TIME OF SENTENCES

	Sentence	Response Time (s)	Accuracy (%)
1.	Hello	1 s	93%
2.	How are you?	1.3 s	94%
3.	Listen up!	1.1 s	92%
4.	I need water.	1.1 s	93%
5.	Please help.	1 s	93%
6.	Do you have a pen?	1.5 s	92%
7.	I need food.	1.2 s	91%
8.	I can do it.	1.2 s	94%
9.	Thank you.	1 s	93%
10.	Switch off the lights.	1.5 s	91%
11.	I am tired.	1.1 s	90%
12.	Feeling accomplished.	1.1 s	91%
13.	Where is the restroom?	1.3 s	93%
14.	Let's go.	1 s	93%

The response time was in the range of 1-1.5 s. It took less time for small sentences and comparatively more seconds for a bit longer one to get converted. Figure 7 shows the accuracy of 14 sentences enlisted in Table I.

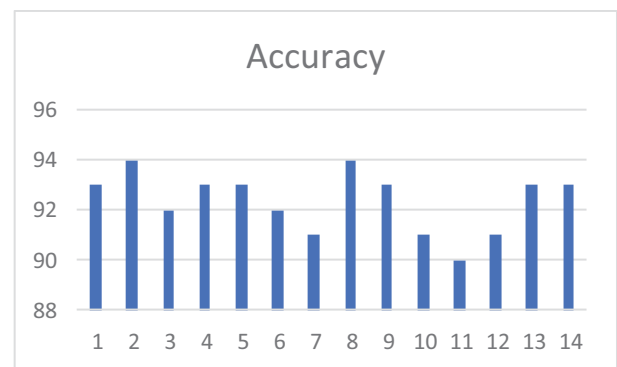


Fig. 7. Bar chart showing accuracy

VI. CONCLUSION

Developing a glove-based Sign Language translator is indeed very helpful in improving the communication between a normal person and a deaf/dumb person. The in-depth review of various previous works reveals the challenges, limitation, benefits and the recommendations for developing a gesture-based communication system. The major advantage here is the sensors directly give the bend value to the glove and the output is available in a fraction of seconds. Also, more and more gestures can

be added into the programming to make up various sentences. Machine learning and artificial intelligence can also be incorporated to increase the accuracy and output delivering capacity of the project. What happens, in that case, is the application starts learning the new gestures rather than categorize them as not legitimate. The proposed framework is a general system as it involves only a microcontroller and an android phone as the primary components. There is no need of any pen or paper for a deaf/dumb individual to convey their message. This framework provides a really easy and simple way to communicate.

Right now, develop a system that can read as many as 23 signs were able to be developed. Similarly, the two gloves can be used to increase the number of gestures. Various gestures can be combined too to increase the sentences. In future, similar projects should take into consideration the weight and size of the system, to make them easy to wear and that they do not involve any interference from a PC or any similar big system. Also, movement of eyes and lips can be incorporated by using various other sensors. The purpose of this particular project was to check the feasibility of recognizing the gestures of various sign languages using sensory gloves. Apart from that, the accuracy of translation can be increased by increasing the number of sensors.

VII. FUTURE WORK

Sensors can be installed to perceive full communication via gestures with more flawlessness and exactness.

The framework can likewise be structured with the end goal that it can interpret words starting with one language then into the next. Improving the proposed framework is a tentative arrangement, and it tends to be finished by first taking care of the issues experienced, and afterwards by including some additional highlights. For instance, auto-adjustment can be accomplished by including another motion which will be intended to eradicate an off-base letter. Different motions can be added to configuration ordinarily utilized expressions, or a point. A similar framework could be additionally executed utilizing other letter set (for example Arabic Alphabet) to suit various societies. Usually, the signs of the sign languages (be it ASL or SL or any other SL) involve two hands and some even include facial expressions to depict any emotion. So, one of the big extensions of this project can be the use of sensors or cameras to capture the facial expressions as well.

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